PROBABILITY LADDER 1 OF 9

Sample spaces and equally likely outcomes

Count the outcomes in an event and divide by all the outcomes, and combine events by adding and complementing.

1 / THE RULE AND THE PICTURE

With equally likely outcomes, $P(A)$ is the count of outcomes in A over the total count

first die \ second die	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

each cell is the sum, and the six gold cells are the sum 7

Two dice have a sample space of $6 \times 6 = 36$ equally likely outcomes, one for each cell. The event "the sum is 7" contains 6 of them, so

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Every probability lies between 0 (an impossible event) and 1 (a certain one), and the probabilities of all outcomes add to 1.

2 / COUNTING EVENTS / A TABLE

Each event is a set of cells, and its probability is the count over 36

event	outcomes	probability	decimal
sum is 7	6/36	1/6	0.167
doubles	6/36	1/6	0.167
sum at least 10	6/36	1/6	0.167
at least one six	11/36	11/36	0.306
both even	9/36	1/4	0.250
sum is prime	15/36	5/12	0.417

The probabilities of the sums $2, \ldots, 12$ rise and fall as $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ over 36, because the sum 7 can be made six ways and the sums 2 and 12 only one.

3 / THE SUMS / A BAR CHART

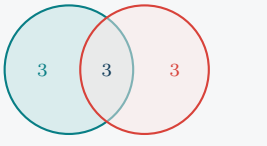
The sum 7 is the most likely, since it has the most cells



The height of each bar counts the cells with that sum, and dividing by 36 gives the probability, for example $P(\text{sum} = 4) = \frac{3}{36} = \frac{1}{12}$. The tempting guess that all sums are equally likely fails, since the outcomes and not the sums are the equally likely things.

4 / COMBINING EVENTS / ADDITION AND COMPLEMENT

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$, and $P(\text{not } A) = 1 - P(A)$



A: sum ≥ 10 B: first is 6

Let A be "sum at least 10" and B "first die is 6". Each has 6 outcomes and they share 3, namely $(6,4)$, $(6,5)$, $(6,6)$, so $P(A \text{ or } B) = \frac{6}{36} + \frac{6}{36} - \frac{3}{36} = \frac{9}{36} = \frac{1}{4}$, which counts 9 distinct cells.
The complement is quicker for "at least one six": $1 - \frac{25}{36} = \frac{11}{36}$, since 25 outcomes show no six.

5 / PROBABILITY AND FREQUENCY

Long runs of trials settle near the probability

In a seeded simulation of 6000 rolls of two dice the sum was 7 on 1001 occasions, a relative frequency of 0.167 against the exact 0.167, and the gap shrinks as the number of trials grows.
Probability is the limit this frequency approaches, and the next sheets show how fast it does.

ASIDE / EXCLUSIVE EVENTS

Events that cannot both happen, such as sum 7 and sum 8, simply add: $\frac{6}{36} + \frac{5}{36} = \frac{11}{36}$.

LINKS TO OTHER SHEETS

Counting without listing is in the next sheet, *Counting: arrangements and choices*; sums of dice return in *Random variables, expectation and variance*.

Convention. S is the sample space; an event is a subset of S; P is probability; the dice are fair and independent.